

Screen Sketches

Group: 4_Kabir_3

Anthony Nehring, Michael Becker, Nicholas Jacobs

Actors

1. **Editors** : have the ability to:

- Change/update personal information such as:
 - Biometric data (height, weight)
 - Username
 - Email address
- View their previous workouts, or track a new one.
- Create a workout template from scratch or from using previous workouts.
- Change profile settings/preferences such as:
 - Notifications
 - Change profile visibility settings to:
 - private- only they can view
 - public- anyone can view
 - friends- only added friends can view
 - Preferred units of measurement

2. **Viewers**: if another user profile is visible to the viewer, then that user's visibility setting allows others to view their profile. A viewer can:

- Look at other user's stats, workout feed (if public).
- Save another user's workout as a "template".
- View leaderboards

3. **Administrators**: have the ability to:

- Manage user posts that have been flagged
- Manage exercise and food libraries
- Create/edit public workout plans that can serve as a template for users.

Non-Functional Requirements

- The application must take no longer than 5 seconds to login and display the user's home page.
- The ui should take no longer than half a second to respond.
- The app should be able to handle many users simultaneously. For the sake of development and considering our resource limitations, it must be able to support 15 users at one time.
- The application must be designed for scalability - in order to support as many users as possible.
- The app must be designed to support modern android devices of various display sizes.

Tables and Fields - Nicholas

User - Stores information related to the user account.

- User ID - Primary Key (System generated)
- First Name
- Last Name
- Email
- Username - name displayed on their profile(unique)
- User Password - use hashing for security
- Last Login date

User Templates- Stores workouts either created or saved by the user, to track workouts.

- Template ID - Primary Key (System generated)
- User ID - Foreign Key
- Date Created
- Workout - stored as JSON object consisting of:
 - Each exercise name
 - Number of sets
 - Number of repetitions per set
 - Planned weight

User Workouts-

- Workout ID - Primary Key (System generated)
- User ID - Foreign Key
- Start Time - (datetime)
- Workout Length
- Total Weight - total weight lifted during the workout.
- Exercises Performed - stored as JSON object consisting of:
 - Each exercise name
 - Number of sets
 - Number of repetitions performed in each set
 - Weight lifted
 - Rate of perceived exertion (RPE)
 - Any additional notes

User Bests- Store each user's personal bests (most weight) of each exercise, at a specific number of repetitions.

- Best ID - Primary Key (System generated)
- User ID - Foreign Key
- Exercise Name
- Exercise Weight
- Exercise Repetitions

User Friends- Stores the many to many relationships of users, tracking each of a users friendships

- Friendship ID - Primary Key (System generated)
- User ID 1 - Foreign Key
- User ID 2 - Foreign Key

User Settings-

- Settings ID - Primary Key (System generated)
- User ID - Foreign Key
- Page Visibility - enum(public, friends, private)
- Biometric Visibility - enum(public, friends, private)
- Weight Unit- enum(lbs, kg)
- Measurement Unit - enum(metric, imperial)
- Message Notifications - bool
- Friend Request Notifications - bool
- Reminder Notifications - bool

User Nutrition Needs-

- Needs ID - Primary Key (System generated)
- User ID - Foreign Key
- Goal Weight- nullable (if null, will determine needs off maintaining current weight)
- Calories
- Protein
- Carbs
- Fats
- Sugar
- Saturated Fats

User Meals -

- Meal ID - Primary Key (System generated)
- User ID - Foreign Key
- Meal type - enum(breakfast, lunch, dinner, snack)
- Meal Date
- Food name
- Serving size
- Servings Consumed

Exercises - Stores known exercises and related info. Using exercise information from open source below, to populate the table by reading a .json file. All users should be able to access this data, as it will be used for tracking the exercises within user workouts. Uses data from repository at <https://github.com/exercemus/exercises>, which is stored in a .json file

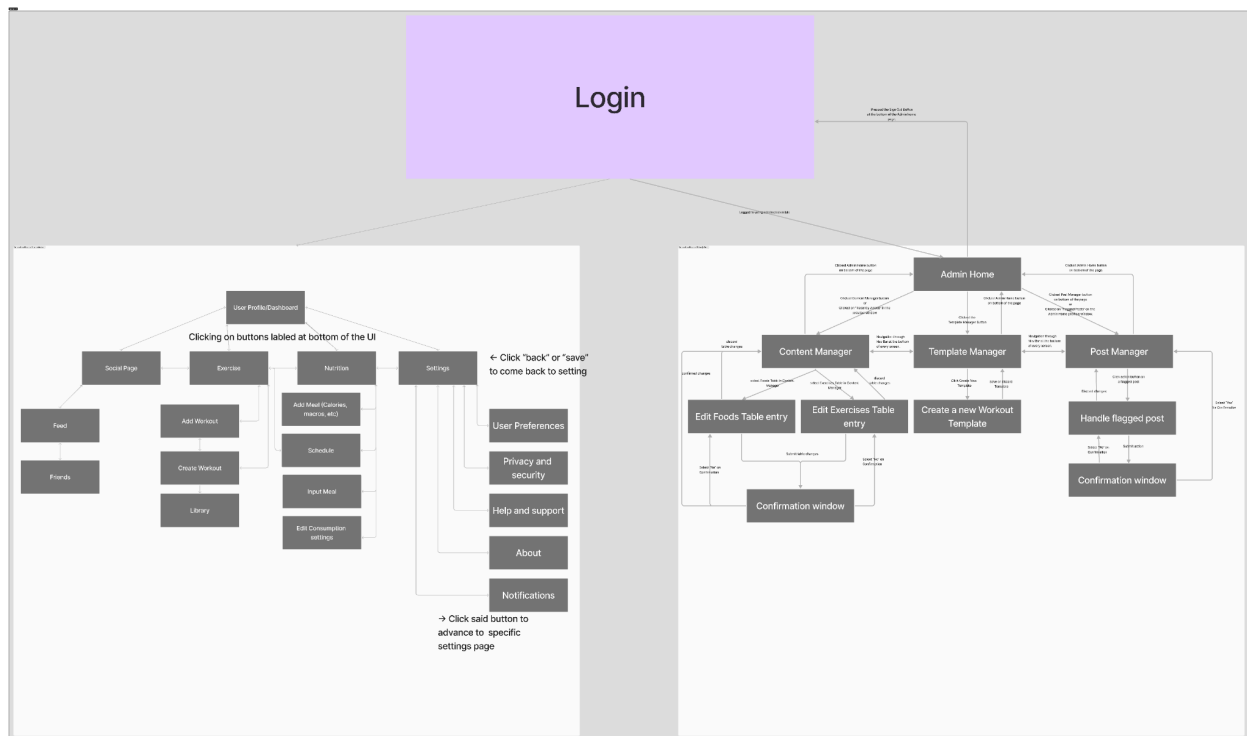
- Exercise ID - Primary Key (System generated)
- Category
- Name
- Aliases - nullable
- Description - nullable
- Instructions
- Tips- nullable
- Equipment
- Primary Muscles
- Secondary Muscles
- Tempo- nullable
- Images - nullable
- Video - nullable
- Variation Of - nullable
- License Author - nullable
- License - nullable

Foods - Not a table that we will have in our DB but we will be pulling from the USDA's open source Food Data Central by using their REST API at:

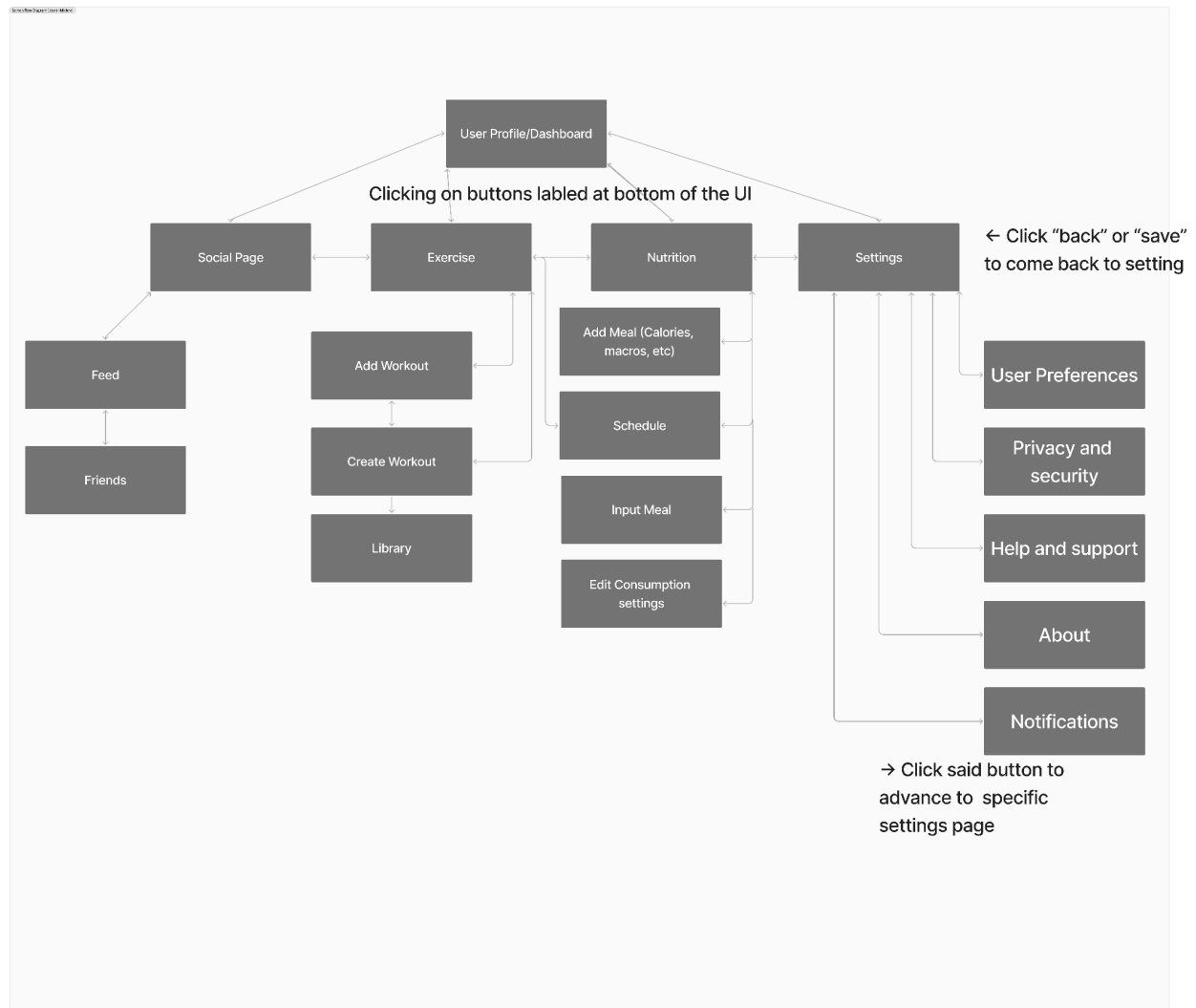
<https://fdc.nal.usda.gov/api-guide.html>

Screen Flow Diagram (Michael)

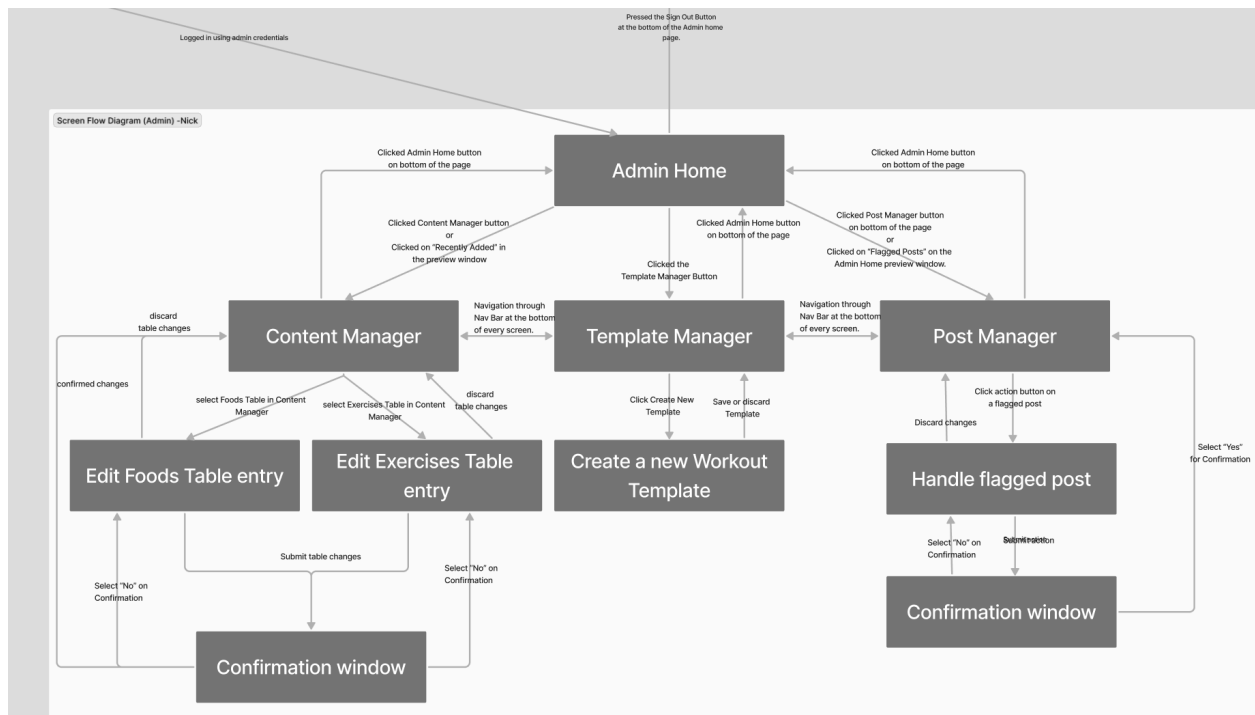
Overall Flow



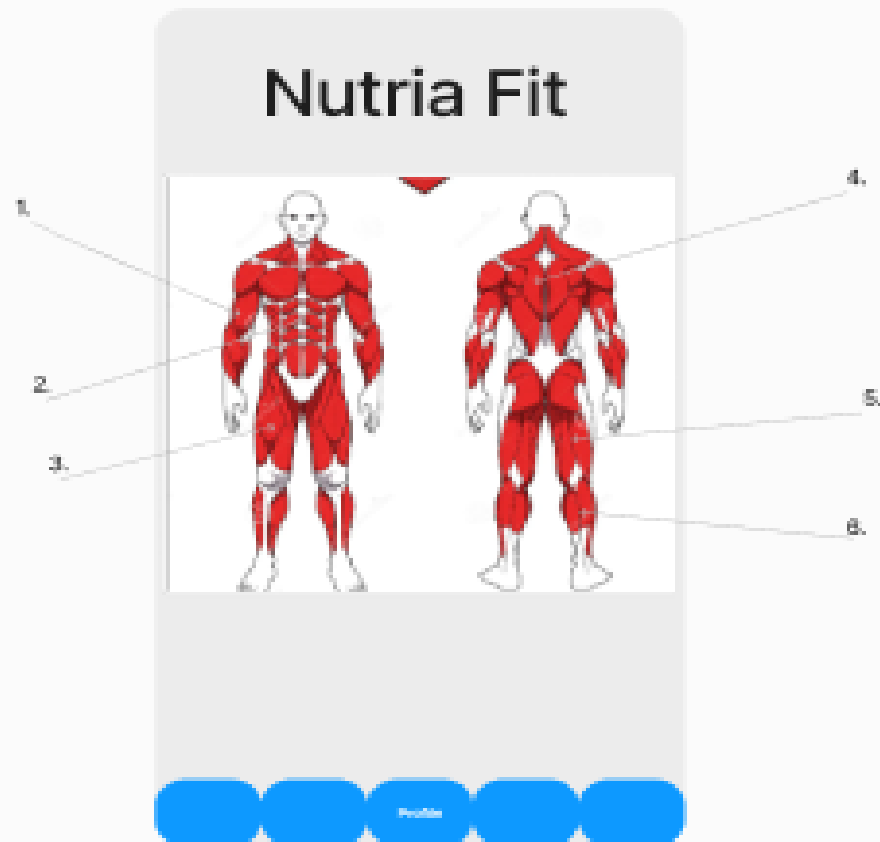
User Flow - Michael



Admin Flow - Nicholas



Home screen/User profile - Anth...



The purpose of the Home screen/user profile is to give the user an interactive image shown above. This screen will allow the user to click on different main muscle groups and be brought to separate screens for each.

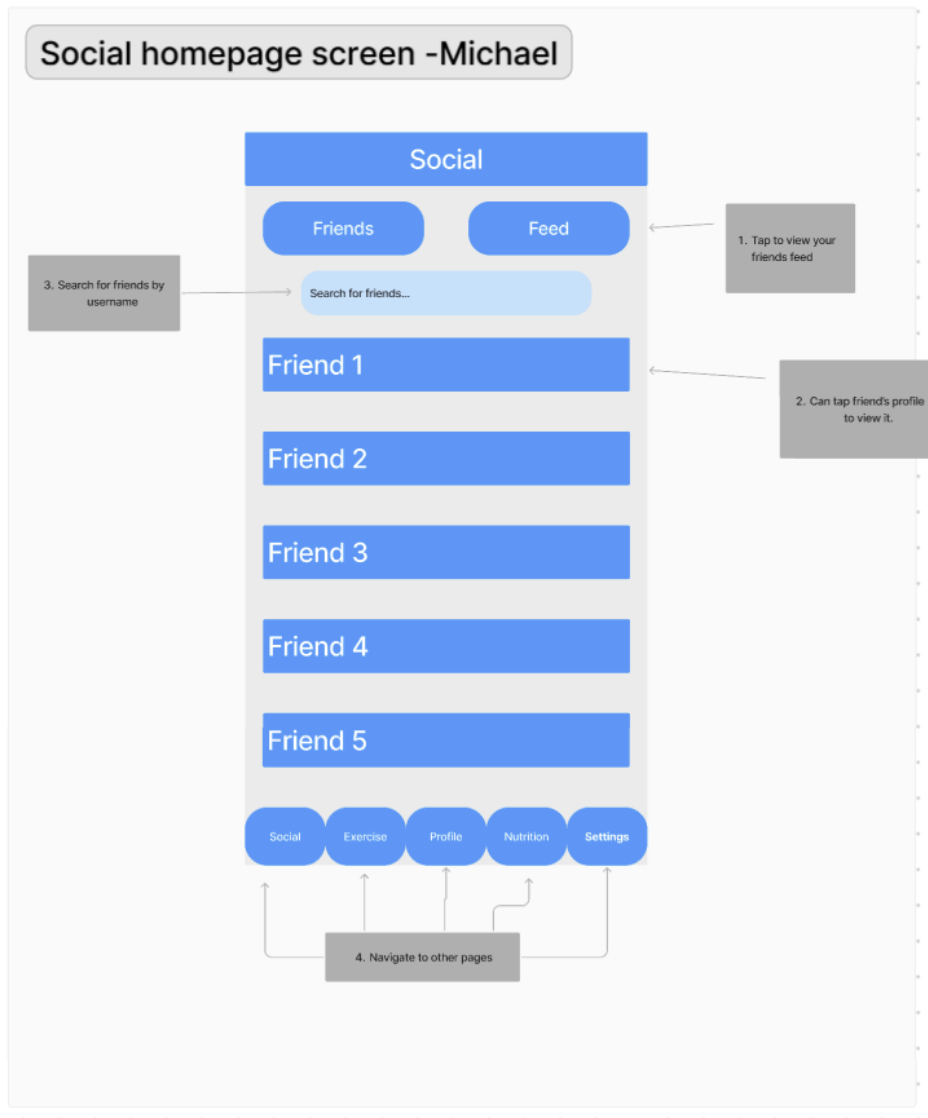
1. bicep page
2. Abs and obliques page
3. Quadriceps page
4. Back page
5. Hamstring page
6. Calf page

Exercise (Michael)



1. Clicking this button will allow you to create a workout of your choosing with a selection of muscle group sets and reps also, including cardio, which will allow you to add to your schedule
2. Allows you to select from a number of different muscle groups to comprise your workout, will take you to a separate page with premade workouts that you can choose and add to your profile/exercises
3. These are your workouts, a list of workouts that have already been made by the user
4. These are favorited workouts that simply allow the user to choose their favorite workouts currently and give them ease of access to add them to their schedule

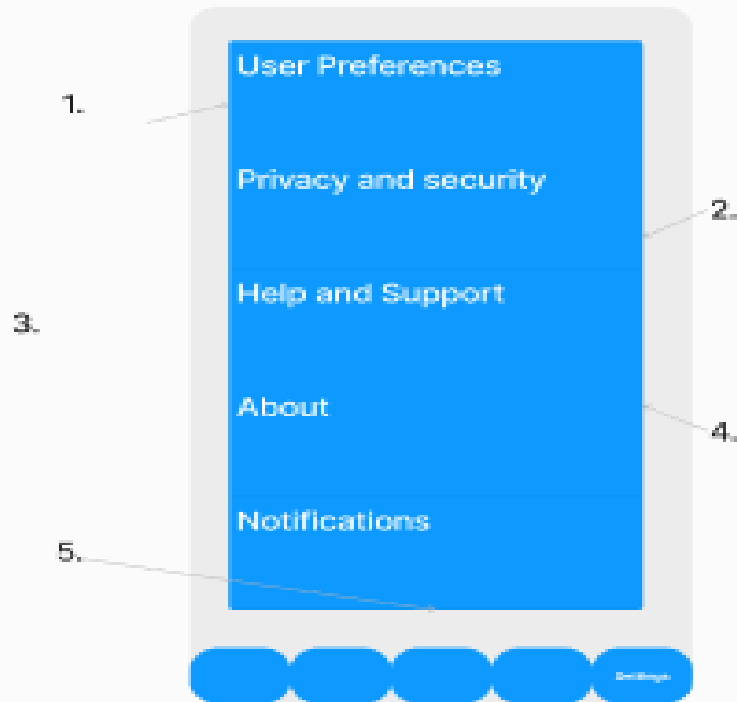
Social Page (Michael)



1. You can tap to see your friend's posts and achievements
2. This allows you to tap your friend's profile and view their profile
3. Pretty self-explanatory, searching for friends by the username they choose when they sign up
4. Allows the user to navigate anywhere on the app with the buttons down below.

Settings - Anthony Nehring

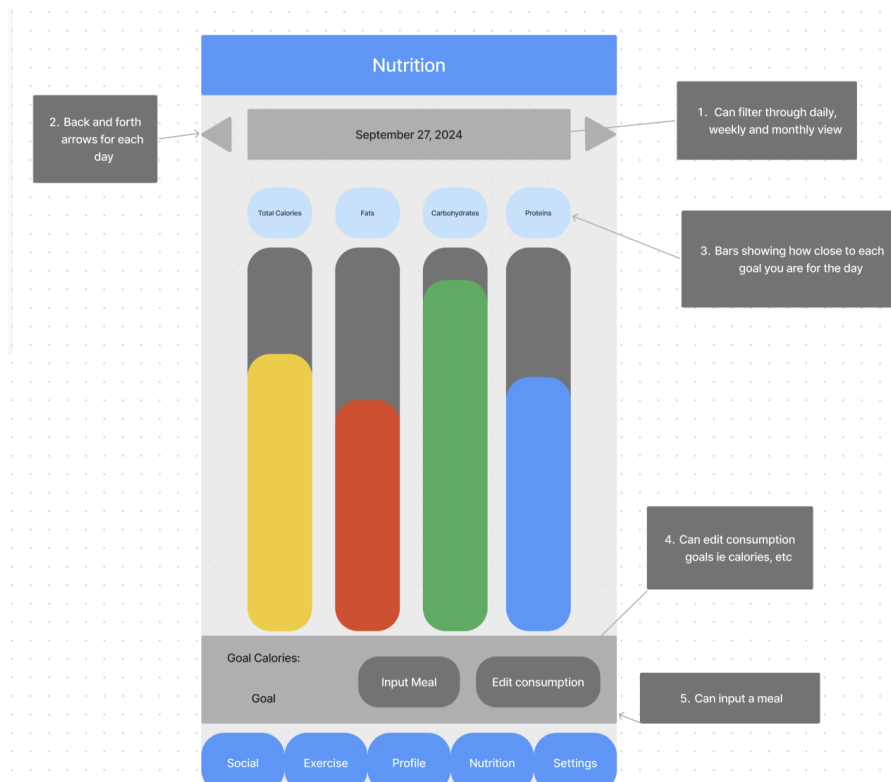
Settings screen - Anthony Nehring



The purpose of the settings screen is to allow users to customize and improve their experience within the app. The settings page implements this through five main buttons that send users into their own separate pages.

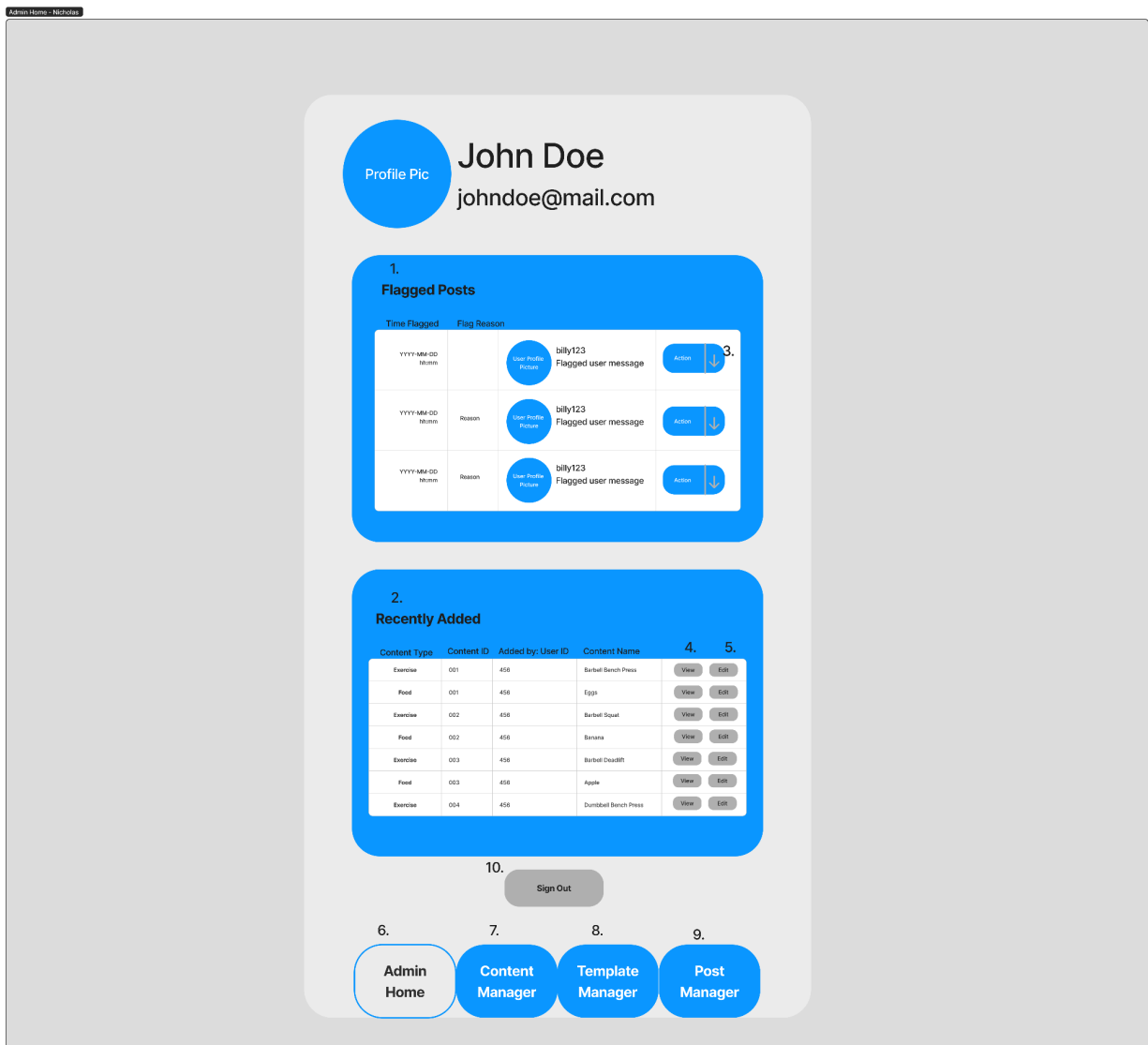
1. User preference - This button will bring you to a page dedicated to user preferences. Implementing "dark/light mode" and etc.
2. privacy and security - This button will bring you to a new page to update password,
3. help and support - This button will bring you to a chat bot - AI implemented or a ticket service.
4. about - This button will open a page that describes what the app is/does
5. notifications - This button will bring you to a page to check yes or no on various notifications that may be revived for this app.

Nutrition (Michael)



1. There will be a dropdown to view your schedule and select daily, weekly, and monthly views for caloric goals as well as fitness goals that have been created.
2. Allows the user to filter through days and see their eating habits, which is useful if A user wants to look back on what they previously ate calories-wise.
3. These bars simply show how close you are to fulfilling your goal, in the final plan, I hope to create percentages to be displayed within the bars themselves comparatively to your goal
4. Edit Consumption button will take you to a page to change your caloric goals
5. Input Meal button will allow the user to add a meal, drawing upon a meal database to help autofill within the categories of macronutrients.

Admin Home - Nicholas

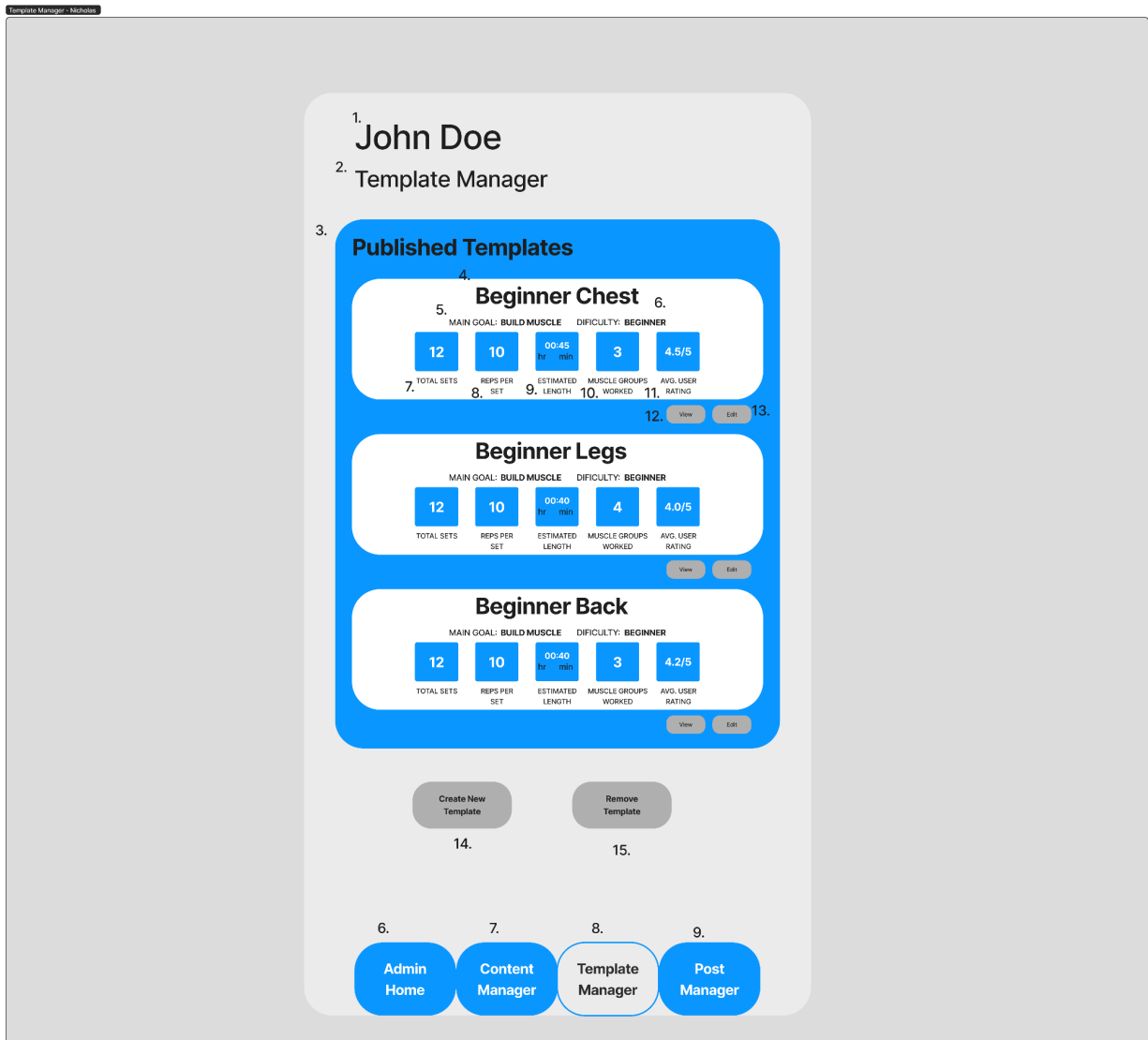


The Admin Home acts as a landing page for admins and allows them to view recent activity that has happened, as well as.

1. Feed of recently flagged posts. Can be clicked on the title to be directed to the Post Manager page instead of using the navigation buttons on the bottom.
2. Feed of recently added entries to the Exercises and Foods tables. Can be clicked on the title to be directed to the content manager page instead of using the navigation buttons on the bottom.
3. Dropdown button to select how to handle the flagged post. Will respond with a popup asking for confirmation of selection and request the admin to log a reason for his decision.

4. Allows the admin to view the full database entry for that item.
5. Allows the admin to edit the database entry for that item inside the content manager.
6. Button returning the admin to their home page.
7. Button taking the admin to the content manager page.
8. Button taking the admin to the template manager page. Template manager allows the admin to create/edit public workout templates that are available to all users.
9. Button taking the admin to the post manager page. Post manager is just an extended version of what is on this screen, allowing the admin to view more posts and not require a popup to complete the action.
10. The button allows the admin to sign out, returning them to the login page.

Template Manager - Nicholas



1. Name of the Admin currently logged in.
2. Name of the page the admin is currently viewing.
3. List of published workout templates available to all users. These can be created for various fitness levels and goals, and make it so the user does not have to make one themselves if they do not want to. The Admin can scroll to view more workouts.
4. Name of the workout template.

5. Fitness Goal of the workout template.
6. Difficulty level of the workout. Range includes Beginner, Intermediate, Advanced, and Expert.
7. The total number of sets planned in this workout.
8. The number of repetitions per set planned in this workout.
9. The estimated length of time to complete the workout.
10. The number of different muscle groups worked during the workout.
11. The average user rating of the workout on a scale of 0 to 5.
12. Button to view that specific workout template. All planned exercises as well as their planned number of repetitions and weight.
13. The button allows the admin to edit that workout template.
14. The button allows the admin to create a new workout template from scratch.
15. Allows the admin to select and then remove a workout template. Will prompt the admin for confirmation.

Create workout template - Anthony

create workout template - Anthony Nehring



1. Add exercise one - by clicking on this button the user will be able to add and implement the first exercise of the workout. The new page/pop up will have things like sets, reps, rest time, and etc.
2. Add exercise two - same functionality as the last button.
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- N. These buttons will extend/ be able to be added until the user sees fit.